Applying the principles of Design Thinking to the Intensive Care Environment

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Patient-centered care and evidence-based medicine are mantras that form the foundation of modern healthcare practice. Yet, most of the tools for designing healthcare solutions, including randomized control trials, quality improvement initiatives, and even qualitative research, are often better adapted to providing an evidence-based foundation for practice at the expense of patient, family, and healthcare worker needs. These approaches tend to focus on improving outcomes and processes (Grys, 2022). Stakeholders — particularly patients, but also staff – are objects being scrutinized by "experts" under the scientific or medical gaze, filtering out what the "expert" deems as irrelevant (O'Callaghan, 2022). However, outcomes and processes are not sufficient if they do not prioritize human voice, dignity, and participation. Design thinking provides a complementary approach to evidence-based medicine by engaging the person in experimenting, prototyping, giving feedback, and redesigning healthcare solutions centered around the needs of humans (Razzouk & Shute, 2012). This article describes the process of design thinking as an approach to the creation of human-centered solutions and makes reference to the implementation of the design thinking process in the intensive care unit (ICU) of SickKids, an academic paediatric hospital in Ontario.

Design thinking originated from the field of engineering, when it became clear that universities were graduating engineers with knowledge of basic sciences, but who lacked the practical skills necessary to work in the industry (Razzouk & Shute, 2012). One can have a sufficient scientific foundation, but lack the ability to solve human-focused problems as people are complex social beings not reducible to deterministic scientific approaches. The engineering curriculum was improved by partnering with real-world users and designing the courses around real problems faced by the end-user (Razzouk & Shute, 2012). Design thinking has now been adopted in various other fields such as business , where it is used to build products and services that address consumer needs more directly (Razzouk & Shute, 2012). Patient-centered medicine demands a focus on both medicine and people; as such, design thinking has the potential to facilitate this person-centered mandate of healthcare systems.

There are many frameworks based on the design thinking process. Examples include the double diamond design popularized by the British Design Council, the Stanford d.school design thinking process, the IDEO Deep-Dive Brainstorming technique, and more. While each framework emphasizes different approaches to the practice of human-centered design, they are all non-linear, non-sequential, and iterative processes that share common traits (Dam & Siang, 2020). This paper focuses on the methodology of the design thinking process, as popularized by Stanford d.school. However, apart from the nomenclature and number of explicitly identified stages, all schools of thought contain similar ideas.

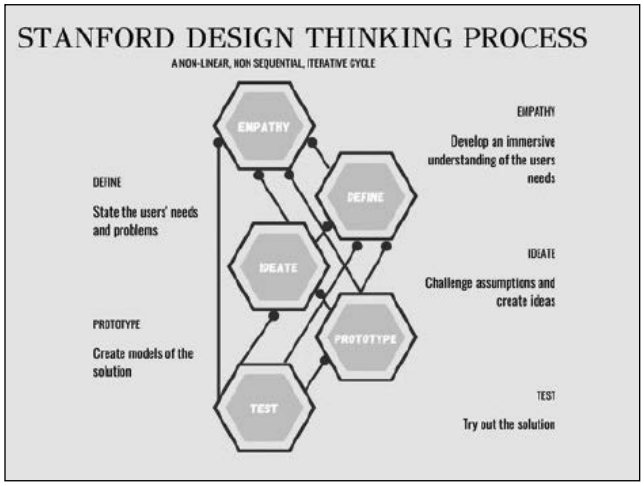


Figure 1. The 5-stage design thinking process of the Stanford design d.school. The stages are empathy, define, ideate, prototype, and test

The Stanford design d.school describes design thinking as a 5-stage process (Figure 1). The stages are "empathy", "define", "ideate", "prototype", and "test" (Dam & Siang, 2020). Empathy is the foundation of design thinking upon which all stages of the process are built, and demands an immersive understanding of the users' needs. It is the act of placing oneself in another person's position and adopting their perspective (Köppen & Meinel, 2015). In practice, this may involve extensive user interviews; observation; participation in, and simulation of, the user experience; and co-designing with the user. For example, in the development of a new ICU at SickKids, designers interviewed nursing stakeholders on their reasoning behind how they organize equipment and supplies at the bedside, and collected hours of video recordings from the ICU to understand how they navigate and perform tasks in said environment. They then constructed large cardboard models of a proposed future ICU environment and invited these stakeholders to perform their

daily activities within these models to replicate the experience of a real ICU, gathering feedback to guide future design iterations. By building empathy and immersing themselves in the experience of the user, designers can build more human-centric solutions.

All stages must be user-centric and defined by empathy. For example, the second stage, "define", works to delineate the users' needs and problems (Dam & Siang, 2020). While the design process often starts with end-users presenting complex problems that demand innovative solutions, the designer must first approach the question with the understanding that the problem may not be well understood to begin with; the end-user-defined presenting problem may merely be a downstream manifestation of a more unifying core problem. For instance, there were concerns about the technique used to provide compressions during open-chest cardiopulmonary resuscitation (CPR) at the aforementioned paediatric ICU of SickKids. By engaging in empathetic interviewing, it was discovered that the problem was not a lack of knowledge about chest compression technique during open-chest CPR, but rather the nurses' anxieties that the patient may get hurt during the process. As a result, the defined problem shifted from being solely education-focused and was broadened to include end-user feelings of anxiety and discomfort with performing the procedure (i.e. CPR). Designing effective solutions is predicated upon an accurate understanding of the key problem(s) and therefore must incorporate an empathetic approach to the end-user, regardless of how confident a designer may initially feel about their understanding of the problem.

After defining a problem, the next stage is "ideation". Ideation is the process of challenging assumptions and creating ideas (Dam & Siang, 2020). One example of a helpful prompt to generate ideas is "how might we" statements (Gottlieb, et al., 2017). These are often framed as, "'How might we' design, implement, develop, change, or et cetera?" This should be focused and based on what was learned from empathetic interviews and processes with end users, and incorporate the defined understanding of the problem. This stage should ideally be done with the end user to generate as many creative ideas as possible, and should strive to generate a large and diverse set of potential solutions. Once ideas are generated, they can be organized into common themes to guide the "prototyping" portion of the design thinking process.

"Prototype" involves creating low-fidelity models of a proposed solution based on the ideas generated during the "ideation" stage. They are often built by working with people from different industries, and the user, in a co-design process. For example, the cardiopulmonary arrest review team at the ICU of the paediatric hospital recognized that the Pediatric Advanced Life Support (PALS) guidelines, as is often the case with evidence-based medicine frameworks, were provider-centric and focused on chest compression rate, depth, and fraction as well as other process-based outcomes as a measure of CPR quality. Patient-centered measures include diastolic blood pressure, end-tidal carbon dioxide (ETCO2), and other physiologic outcomes of the patient (Lin & Scales, 2016), and are being increasingly described within the resuscitation literature. The review team ideated by asking the question, "'How might we' build a data visualization system that will correlate both provider-centric and patient-physiologic responses to resuscitation efforts to understand and improve resuscitation quality?" In this case, the prototype was built by working with computer scientists and programmers to incorporate chest compression data from defibrillators and patient physiologic arterial line and ETCO2 waveforms into an application that correlates chest compression measures with patient-physiologic responses, thus providing novel insights into resuscitation quality.

Once a working prototype has been created, the design process moves onto the "test" phase. In this stage, the end user tries out the solution using either simulation or in-situ testing (Dam & Siang, 2020). In reference to the previous example of the cardboard models for the new ICU building, stakeholders such as nurses and patients are brought into the model, and experiences of routine interactions in the ICU are simulated. It is important to receive as harsh of a critique from the stakeholders as possible to ensure that the solutions fit the needs of the users. It is also important to remember that every solution is at best a temporary one, even after real-world implementation. The design thinking process is iterative and is always questioning and observing how people and their needs change over time in order to adapt new solutions.

In conclusion, design thinking offers a standardized framework for fulfilling the mandate of person-centered medicine. This article has explored the stages of one of the dominant approaches to the design thinking process using the experiences of an ICU in SickKids, a paediatric hospital in Ontario. These stages are non-linear, non-sequential, and iterative, with empathy and co-design being the foundation of all other stages. Design thinking methodology is already used to great effect in a variety of industries. Medicine and the people it serves benefit from a framework that is adapted to person-centered care.

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